

EEB1323-MICROPROCESSOR AND MICROCONTROLLER

SMART AUTOMATED MEDICINE DISPENSING AND ALERT

SYSTEM FOR BLIND AND OLD AGE PEOPLE

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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PSO3: Design, Develop and implement methods and concepts to facilitate solutions for electrical and electronics engineering-related real-world problems.

Abstract (Key words)	POs Mapping
The Smart Medicine Dispensing and Alert System helps old and blind people take medicine on time. It stores tablets in a cylinder box. At the set time, it gives the tablet and makes a buzzer sound. It also sends a message to the caretaker if the medicine is taken or not.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10 PO11, PSO1, PSO2, PSO3

SDG MAPPING

SDG N0.	SDG Goal	Remarks
SDG 9	Industry ,Innovation and Infrastructure	This project supports SDG Goal 9 by using smart and affordable technology to help blind and elderly people take their medicines safely and on time
SDG 3	Good Health and Well-being	This project supports SDG Goal 3 by improving healthcare through timely and accurate medication management.

PROFESSIONAL BODY STANDARDS

Professional Body Standards Number	Professional Body Standards Name	Remarks
IEEE 11073	IEEE Standard for Medical Device Communication	This standard supports communication between medical devices and external systems. It ensures reliable data exchange for monitoring patient health and medication intake. It helps in integrating the smart medicine dispenser with healthcare systems for better tracking and safety.
IEEE 802.11	IEEE Standard for Wireless Local Area Networks (Wi-Fi)	This standard enables wireless communication between the device and mobile application. It supports real-time data transmission, remote monitoring, and notification alerts through IoT connectivity. It ensures stable and efficient communication.

ABSTRACT

Medicines play a vital role in maintaining good health, but many elderly and visually impaired individuals face significant challenges in managing their daily medication schedules. Forgetting to take medicines on time or consuming them at incorrect intervals can lead to serious health complications, reduced treatment effectiveness, and increased dependency on caregivers. To address these issues, a Smart Automated Medicine Dispensing and Alert System is developed to ensure that users take the correct medicine at the right time without relying on memory or constant supervision. The proposed system is based on an IoT-enabled microcontroller that connects with a mobile application to allow users or caregivers to set medication schedules such as morning, afternoon, and night. Medicines are stored in a cylindrical compartment system designed for organized and automatic dispensing. At the scheduled time, a servo motor rotates the cylinder to align the correct compartment and releases the required tablet. A buzzer alert is activated to notify the user, ensuring that the medicine is not missed. Additionally, the system includes a monitoring mechanism to detect whether the medicine has been taken. If the tablet is consumed, a confirmation notification is sent to the caregiver. If the medicine is not taken within the specified time, an alert message is sent through the IoT module. This feature enables real-time monitoring and enhances patient safety. Overall, the Smart Medicine Dispensing System is a simple, cost-effective, and user-friendly solution that improves medication adherence, reduces human errors, and supports independent living for elderly and visually impaired individuals.

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CHAPTER 1

INTRODUCTION

In today's fast-paced lifestyle, maintaining a proper medication schedule has become a major challenge, particularly for elderly and visually impaired individuals. Many patients forget to take their medicines on time, take incorrect dosages, or confuse different medications, which can lead to serious health complications and reduced effectiveness of treatment. Traditional methods such as handwritten schedules, basic pill boxes, and manual reminders are not always reliable, as they depend heavily on the user's memory and continuous support from caregivers [3], [10].

To address these issues, recent advancements in technology have introduced smart healthcare systems that utilize IoT and embedded systems for better medication management. IoT-based solutions enable real-time monitoring, automated reminders, and improved communication between patients and caregivers, thereby enhancing patient safety and treatment adherence [1], [9]. Systems that incorporate GSM or Wi-Fi modules can send notifications and alerts to caregivers, allowing them to monitor medication intake even from remote locations, reducing the need for constant physical supervision [3], [8].

Furthermore, modern smart pill dispensers are equipped with sensors that can detect whether the medicine has been taken or not, providing accurate feedback and ensuring proper medication tracking [5], [7]. Some systems also include voice assistance features to support visually impaired users, making them more accessible and user-friendly [2], [12]. These advancements significantly improve the usability and effectiveness of automated medication systems.

Based on these developments, the proposed Smart Automated Medicine Dispensing and Alert System aims to provide a reliable and efficient solution for medication management. The system uses IoT technology to automate the dispensing process, provide timely alerts, and enable real-time monitoring through a mobile

application. It ensures that the right medicine is taken at the right time while reducing dependency on caregivers. This approach enhances patient safety, improves medication adherence, and supports independent living for elderly and visually impaired individuals [4], [6].

1.1 OBJECTIVE OVERVIEW

The main objective of this project is to design and develop a Smart Automated Medicine Dispensing and Alert System that ensures patients take the correct medicine at the correct time without confusion. The system aims to automate the medicine dispensing process using a cylindrical storage mechanism controlled by a servo motor, reducing the need for manual intervention.

Another important objective is to provide timely alerts to users through a buzzer, helping them remember to take their medicine regularly. The system also focuses on monitoring whether the medicine is taken or not using sensors, ensuring accurate tracking of medication intake. The project further aims to implement IoT technology for real-time communication, allowing caregivers to receive notifications about medicine intake or missed doses through a mobile application. This enhances patient safety and enables remote monitoring. The project further aims to implement IoT technology for real-time communication, allowing caregivers to receive notifications about medicine intake or missed doses through a mobile application. This enhances patient safety and enables remote monitoring. Overall, the objective is to improve medication adherence, reduce human errors, support elderly and visually impaired individuals, and provide a simple, reliable, and cost-effective healthcare solution. Overall, the objective is to improve medication adherence, reduce human errors, support elderly and visually impaired individuals, and provide a simple, reliable, and cost-effective healthcare solution.

1.2 SCOPE OF THE PROJECT

The scope of this project is to develop an IoT-based Smart Automated Medicine Dispensing and Alert System that helps users manage their medication effectively. The system is designed to automate the process of dispensing tablets at scheduled times such as morning, afternoon, and night, reducing the need for manual intervention. It includes the use of a cylindrical storage mechanism and servo motor to ensure accurate and timely medicine delivery. The project also focuses on integrating sensors to detect whether the medicine has been taken, enabling proper monitoring of medication intake. The system allows users or caregivers to set medicine schedules through a mobile application and provides real-time notifications regarding medicine status. This ensures remote monitoring and improves patient safety. The system allows users or caregivers to set medicine schedules through a mobile application and provides real-time notifications regarding medicine status. This ensures remote monitoring and improves patient safety.

The project is mainly intended for elderly and visually impaired individuals, helping them maintain proper medication routines independently. It is designed to be simple, cost-effective, and suitable for home healthcare applications, with the possibility of future upgrades such as voice assistance and cloud data storage.

1.3 PROPOSED SYSTEM

The proposed system is a Smart Automated Medicine Dispensing and Alert System designed to ensure timely and accurate medication intake using IoT technology. The system uses a microcontroller such as ESP32 or Node MCU as the main control unit, which manages all operations including scheduling, dispensing, alerting, and communication. Medicines are stored in a cylindrical container with multiple compartments assigned for different time periods such as morning, afternoon, and night. Medicines are stored in a cylindrical container with multiple compartments assigned for different time periods such as morning, afternoon, and night. The user or caregiver sets the medication schedule through a mobile

application. Based on the set timings, the system automatically activates a servo motor to rotate the cylinder and dispense the correct tablet at the right time.

A buzzer alert is generated to notify the user when it is time to take the medicine. Sensors such as IR sensors or switches are used to detect whether the tablet has been taken. This information is processed by the system to determine the medication status. The IoT module enables real-time communication by sending notifications to caregivers through the mobile application. If the medicine is taken, a confirmation message is sent, and if it is not taken, an alert is generated. The IoT module enables real-time communication by sending notifications to caregivers through the mobile application. If the medicine is taken, a confirmation message is sent, and if it is not taken, an alert is generated.

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CHAPTER 2

LITERATURE REVIEW

1. Paper Title: IOT-Based Smart Medicine Box for Automated Medication Management

Authors: R. Kumar, M. Venkatesh et al

Year: 2021

Description:

This paper presents an IoT-enabled smart medicine box designed to remind elderly and visually impaired users about their medication schedules. The system integrates sensors, a microcontroller, and GSM messaging to provide real-time alerts. The authors highlight how automated reminders reduce missed doses and improve treatment adherence. The authors highlight how automated reminders reduce missed doses and improve treatment adherence. The authors highlight how automated reminders reduce missed doses and improve treatment adherence. The device's ability to send notifications to caregivers ensures additional safety, especially for patients living alone. Experimental results show high reliability and consistent performance during long-term testing.

2. Paper Title: Voice-Assisted Pill Dispenser for Blind Patients Using Microcontroller

Authors: S.Sharma,K.Patel et al

Year:2020

Description:

This work proposes a pill dispenser that uses voice instructions to guide blind users in taking their medicines correctly. The system includes IR sensors for compartment monitoring and a speech module for audio alerts. The study emphasizes accessibility and ease of use for visually impaired individuals. Tests show that voice assistance significantly reduces medication errors. The authors conclude that incorporating speech technology increases independence and confidence among blind users in daily medication routines.

3. Paper Title: Smart Medication Alert System Using GSM and Sensors

Authors:A.Desai,P.Rao et al

Year:2019

Description:

The paper introduces a smart alert system that notifies patients via buzzer and SMS when it is time to take medicine. A combination of IR sensors and a real-time clock module ensures accurate dose timing. When the user opens the medicine tray, the system records the event, helping caregivers track compliance. When the user opens the medicine tray, the system records the event, helping caregivers track compliance. A combination of IR sensors and a real-time clock module ensures accurate dose timing. When the user opens the medicine tray, the system records the event, helping caregivers track compliance. The study reports increased adherence in elderly patients. The authors suggest that low-cost components and simple architecture make the design suitable for large-scale adoption.

4. Paper Title: Automated Pill Organizer with IOT Monitoring for Elderly

Healthcare

Authors:L.Fernandes,R.Joseph et al

Year:2022

Description:

This research presents an automated pill organizer capable of sorting and dispensing medicines at present times. The system uses IOT connectivity to update caregivers on patient activity. The authors demonstrate that automation reduces confusion among elderly individuals who take multiple medicines daily. The system uses IOT connectivity to update caregivers on patient activity. The authors demonstrate that automation reduces confusion among elderly individuals who take multiple medicines daily. The authors demonstrate that automation reduces confusion among elderly individuals who take multiple medicines daily. The design features an intuitive interface and robust alert mechanisms. Realworld testing confirms that the device improves medication accuracy and reduces caregiver burden.

5. Paper Title: Smart Pill Box with Sensor-Based Feedback and Caregiver Notification

Authors:K.S.Anand,B.Babu et al

Year:2023

Description:

This study focuses on a smart pill box equipped with sensors that detect whether the medicine has been taken. If a dose is missed, the system automatically alerts caregivers through a mobile application. The authors highlight the importance of real-time monitoring for patients with chronic illnesses. If a dose is missed, the system automatically alerts caregivers through a mobile application. The authors highlight the importance of real-time monitoring for patients with chronic illnesses. The design integrates sensor data with cloud storage, enabling longterm tracking. Experiments show the system's high accuracy in detecting missed doses and its potential to support remote healthcare.

6. Paper Title: IoT-Integrated Smart Medication Dispenser with Real-Time Monitoring

Authors:N.Verma,S.Raj et al

Year:2023

Description:

This paper introduces a smart medication dispenser that integrates IoT connectivity and sensor-based monitoring to support elderly and visually impaired individuals. The system uses IR sensors, a real-time clock, and a cloud dashboard to track medicine intake and notify caregivers when a dose is missed. The authors highlight the importance of real-time data for improving patient safety and reducing dependence on manual reminders. The authors highlight the importance of real-time data for improving patient safety and reducing dependence on manual reminders. Experimental results demonstrate efficient alert delivery, high sensor accuracy, and strong reliability during long-term use. The study concludes that IoT-driven dispensers can significantly enhance medication adherence and remote healthcare support.

7. Paper Title: Smart Pill Dispenser Using IoT for Healthcare Monitoring

Authors:R.Kumar,P.Singh et al

Year:2022

Description:

This paper presents an IoT-based smart pill dispenser designed to assist patients in maintaining proper medication schedules. The system uses a microcontroller, RTC module, and Wi-Fi connectivity to automate pill dispensing and send notifications to caregivers. It includes buzzer alerts and sensor-based detection to confirm pill intake. The system uses a microcontroller, RTC module, and Wi-Fi connectivity to automate pill dispensing and send notifications to caregivers. It includes buzzer alerts and sensor-based detection to confirm pill intake. The results show improved medication adherence and reduced chances of missed doses, making it suitable for elderly and chronically ill patients.

8. Paper Title: Automated Medicine Dispenser with Remote Monitoring System

Authors:A.Sharma,M.Gupta et al

Year:2021

Description:

This study focuses on an automated medicine dispensing system that uses embedded systems and IoT technology. The device dispenses medicines at scheduled times and provides alerts through buzzer and display. It also sends SMS notifications using a GSM module to inform caregivers about medicine intake. The system improves reliability and ensures patient safety by reducing manual errors in medication management. This study focuses on an automated medicine dispensing system that uses embedded systems and IoT technology. The device dispenses medicines at scheduled times and provides alerts through buzzer and display. It also sends SMS notifications using a GSM module to inform caregivers about medicine intake. The system improves reliability and ensures patient safety by reducing manual errors in medication management.

9. Paper Title: IoT-Based Smart Healthcare System for Medication Management

Authors: S.Iyer, K.Nair et al

Year: 2020

Description:

This paper proposes a smart healthcare system that uses IoT for managing medication schedules. The system includes sensors, a microcontroller, and a cloud platform to monitor medicine intake in real time. Alerts are provided through mobile applications, and caregivers receive notifications for missed doses. The study highlights the effectiveness of IoT in improving healthcare monitoring and patient compliance. This paper proposes a smart healthcare system that uses IoT for managing medication schedules. The system includes sensors, a microcontroller, and a cloud platform to monitor medicine intake in real time. Alerts are provided through mobile applications, and caregivers receive notifications for missed doses. The study highlights the effectiveness of IoT in improving healthcare monitoring and patient compliance.

10. Paper Title: Design of Intelligent Medicine Reminder and Dispenser System

Authors: V.Patel, D.Mehta et al

Year: 2019

Description:

This paper describes an intelligent medicine reminder system that combines automation and alert mechanisms. The system uses a timer-based control unit, servo motors, and sensors to dispense medicines and detect usage. It provides audio alerts and supports remote monitoring through wireless communication. The results indicate that the system is cost-effective, user-friendly, and beneficial for elderly users. This paper describes an intelligent medicine reminder system that combines automation and alert mechanisms. The system uses a timer-based control unit, servo motors, and sensors to dispense medicines and detect usage. It provides audio alerts and supports remote monitoring through wireless communication. The results indicate that the system is cost-effective, user-friendly, and beneficial for elderly users.

CHAPTER 3

EXISTING SYSTEM

3.1 INTRODUCTION

Existing medication management systems primarily rely on traditional and manual methods such as pill organizers, written schedules, and basic alarm reminders. These approaches depend heavily on the user's memory, attention, and discipline, which can be unreliable, especially for elderly individuals, visually impaired users, and patients with memory-related issues. As a result, users may forget to take their medicines, take them at the wrong time, or even consume incorrect dosages, leading to serious health complications and reduced effectiveness of treatment.

Some modern systems include simple electronic reminders using alarms or timers to notify users when it is time to take medicine. However, these systems only provide alerts and do not ensure whether the medicine has actually been taken. There is no proper mechanism to verify medication intake or track user response. This lack of feedback makes these systems less effective in ensuring proper adherence to medication schedules. In addition, most existing systems do not include real-time monitoring or communication features. Caregivers and family members are not informed about whether the patient has taken the medicine or missed a dose. This becomes a major limitation, especially for patients who live alone or require constant supervision. Caregivers often have to rely on manual checking, which is time-consuming and not always possible due to distance or other responsibilities.

Furthermore, existing systems lack automation and intelligent control. They do not have the capability to dispense medicine automatically at the correct time, increasing the chances of human error. Without proper integration of sensors, IoT technology, and feedback mechanisms, these systems fail to provide a complete and reliable solution for medication management. Overall, the limitations of existing systems—such as lack of automation, absence of monitoring, and dependency on human memory—highlight the need for a smart and efficient solution that can ensure.

3.2 Block Diagram

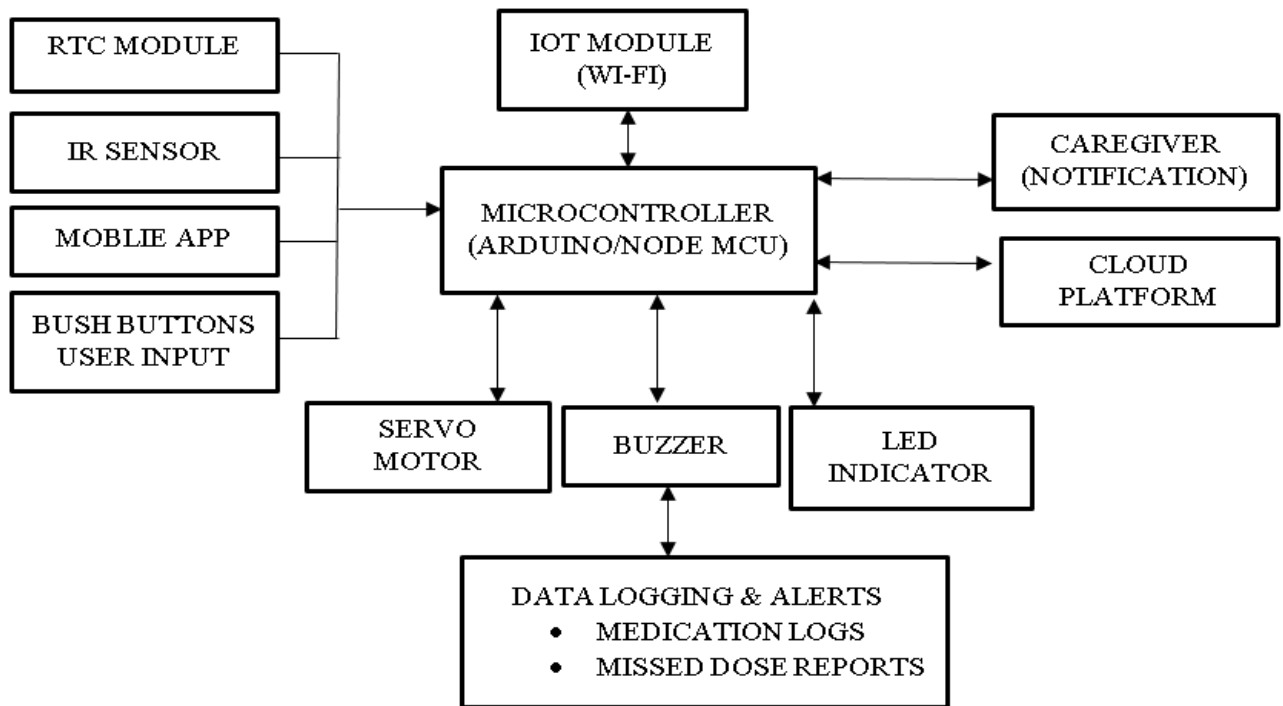


Figure 3.2.1 Existing Block Diagram

The diagram represents the working architecture of the Smart Automated Medicine Dispensing and Alert System, showing how different components interact to ensure proper medication management. At the center of the system is the microcontroller (Arduino/Node MCU), which acts as the main control unit. It receives inputs, processes data, and controls all outputs. On the input side, components such as the RTC module, IR sensor, mobile application, and push buttons provide necessary information to the controller. The RTC module keeps track of real-time, ensuring medicines are dispensed at the correct schedule. The mobile app allows users or caregivers to set timings, while sensors help detect whether the medicine is taken. On the output side, the microcontroller controls devices like the servo motor, buzzer, and LED indicator. The servo motor is responsible for dispensing the tablet by rotating the cylindrical storage. The buzzer gives an audio alert to remind the user, and the LED provides visual indication.

The system also includes an IoT module (Wi-Fi/GSM), which enables communication with the caregiver or user through SMS or app notifications. It also connects to a cloud platform, where data such as medication logs and missed doses are stored and monitored. The system also includes an IoT module (Wi-Fi/GSM), which enables communication with the caregiver or user through SMS or app notifications. It also connects to a cloud platform, where data such as medication logs and missed doses are stored and monitored. Overall, the diagram shows a complete system where sensing, processing, alerting, and communication work together to ensure timely medication and remote monitoring. Overall, the diagram shows a complete system where sensing, processing, alerting, and communication work together to ensure timely medication and remote monitoring.

3.2.1 Power Supply and System Initialization

The system is powered using a stable power supply that provides the required voltage to all components. Once powered ON, the microcontroller initializes all connected modules such as the IoT unit, sensors, and servo motor. The medicine schedule is configured through a mobile application by the user or caregiver, and this data is stored in the system memory for further operation. Proper initialization ensures smooth functioning of the entire system.

3.2.2 Time Monitoring and Scheduling

The system continuously monitors the current time using the IoT module or an internal timer. It compares the real-time clock value with the predefined medication schedule such as morning, afternoon, and night. This continuous comparison ensures that the system can trigger actions at the exact scheduled time without delay, improving accuracy in medication management.

3.2.3 Automatic Tablet Dispensing

When the current time matches the scheduled time, the microcontroller sends a signal to the servo motor. The servo motor rotates the cylindrical storage to align

the correct compartment, allowing the tablet to be dispensed automatically into the collection area. This automation reduces manual effort and eliminates the chances of human error in medicine handling.

3.2.4 Alert and Notification System

After dispensing the tablet, a buzzer is activated to alert the user. The alert continues until the medicine is taken. Sensors such as IR sensors or switches detect whether the tablet has been removed. Based on this detection, the system sends a notification to the caregiver through the IoT module, indicating whether the medicine is taken or missed. This ensures proper monitoring and improves patient safety.

3.2.5 Medicine Detection and Feedback System

The system uses sensors such as IR sensors or limit switches to detect whether the dispensed tablet has been taken by the user. When the tablet is removed from the collection area, the sensor sends a signal to the microcontroller. This feedback helps the system confirm medicine intake and update the status accordingly. It ensures accurate tracking and avoids false alerts.

3.2.6 IoT Communication and Data Transmission

The IoT module (ESP32/Node MCU) enables wireless communication between the device and the mobile application. It sends real-time data such as medicine taken or missed status to the caregiver. This communication ensures remote monitoring and allows caregivers to stay updated about the patient's medication routine from anywhere.

3.2.7 Mobile Application Interface

A user-friendly mobile application is used to set medicine schedules and monitor system status. Users or caregivers can easily configure timing, select compartments, and receive notifications. The application provides a simple interface that improves usability and makes the system accessible to all age groups.

3.2.8 Data Logging and Monitoring

The system stores important data such as medication timings, taken status, and missed doses. This data can be viewed through the mobile app or cloud platform. It helps caregivers track patient behavior and analyze medication adherence over time.

3.2.9 Safety and Error Handling

The system includes basic error handling to manage issues such as motor failure, sensor malfunction, or communication errors. It ensures that the system continues to operate reliably and provides alerts in case of any malfunction, improving overall safety and performance.

3.3 Problem Statement

Medication management is a significant challenge for many individuals, especially elderly people, visually impaired users, and patients with memory-related disorders. They often forget to take medicines on time or may take incorrect dosages, which can lead to serious health complications and reduced treatment effectiveness. Missing doses or improper timing of medication can worsen existing health conditions and increase medical risks. Additionally, traditional methods rely heavily on memory or caregiver support, which may not always be available. Therefore, there is a strong need for a smart and reliable system that can ensure timely medication, reduce human errors, and provide continuous monitoring for better healthcare support.

CHAPTER 4

PROPOSED SYSTEM

4.1 INTRODUCTION

The proposed system aims to develop a Smart Automated Medicine Dispensing and Alert System using IoT technology to ensure timely and accurate medication intake. The system is designed to assist elderly and visually impaired individuals by reducing dependency on memory and manual supervision. It uses a microcontroller such as ESP32 or Node MCU as the main control unit to manage all operations including scheduling, dispensing, alerting, and communication.

Medicines are stored in a cylindrical compartment system, where each section is assigned for specific time periods such as morning, afternoon, and night. The user or caregiver can set the medication schedule through a mobile application, making the system easy to use and flexible. Medicines are stored in a cylindrical compartment system, where each section is assigned for specific time periods such as morning, afternoon, and night. The user or caregiver can set the medication schedule through a mobile application, making the system easy to use and flexible.

At the scheduled time, the system automatically activates a servo motor to rotate the cylinder and dispense the required tablet. A buzzer alert is generated to notify the user, ensuring that the medicine is taken on time. Sensors are used to detect whether the tablet has been taken, providing feedback to the system. At the scheduled time, the system automatically activates a servo motor to rotate the cylinder and dispense the required tablet. A buzzer alert is generated to notify the user, ensuring that the medicine is taken on time. Sensors are used to detect whether the tablet has been taken, providing feedback to the system. The IoT module enables real-time communication by sending notifications to caregivers about the medication status. If the medicine is taken, a confirmation message is sent; if not, an alert is generated.

Overall, the proposed system offers a smart, reliable, and cost-effective solution for improving medication adherence and enhancing healthcare support.

4.2 CIRCUIT DIAGRAM

This circuit diagram represents the complete working of the Smart Automated Medicine Dispensing and Alert System, showing how different hardware components are interconnected. This circuit diagram represents the complete working of the Smart Automated Medicine Dispensing and Alert System, showing how different hardware components are interconnected. At the center of the system is the microcontroller, which acts as the brain and controls all operations. It is powered by a 5V power supply, which also powers other components like the Wi-Fi module, relay module, and sensors.

The RTC (Real-Time Clock) module is connected to the microcontroller to maintain accurate time. It helps the system trigger medicine dispensing at the correct scheduled time. The Wi-Fi module is used for IoT communication, enabling the system to send notifications to caregivers and receive data from the mobile application. The RTC (Real-Time Clock) module is connected to the microcontroller to maintain accurate time. It helps the system trigger medicine dispensing at the correct scheduled time. The Wi-Fi module is used for IoT communication, enabling the system to send notifications to caregivers and receive data from the mobile application. The RTC (Real-Time Clock) module is connected to the microcontroller to maintain accurate time. It helps the system trigger medicine dispensing at the correct scheduled time. The Wi-Fi module is used for IoT communication, enabling the system to send notifications to caregivers and receive data from the mobile application. The Wi-Fi module is used for IoT communication, enabling the system to send notifications to caregivers and receive data from the mobile application.

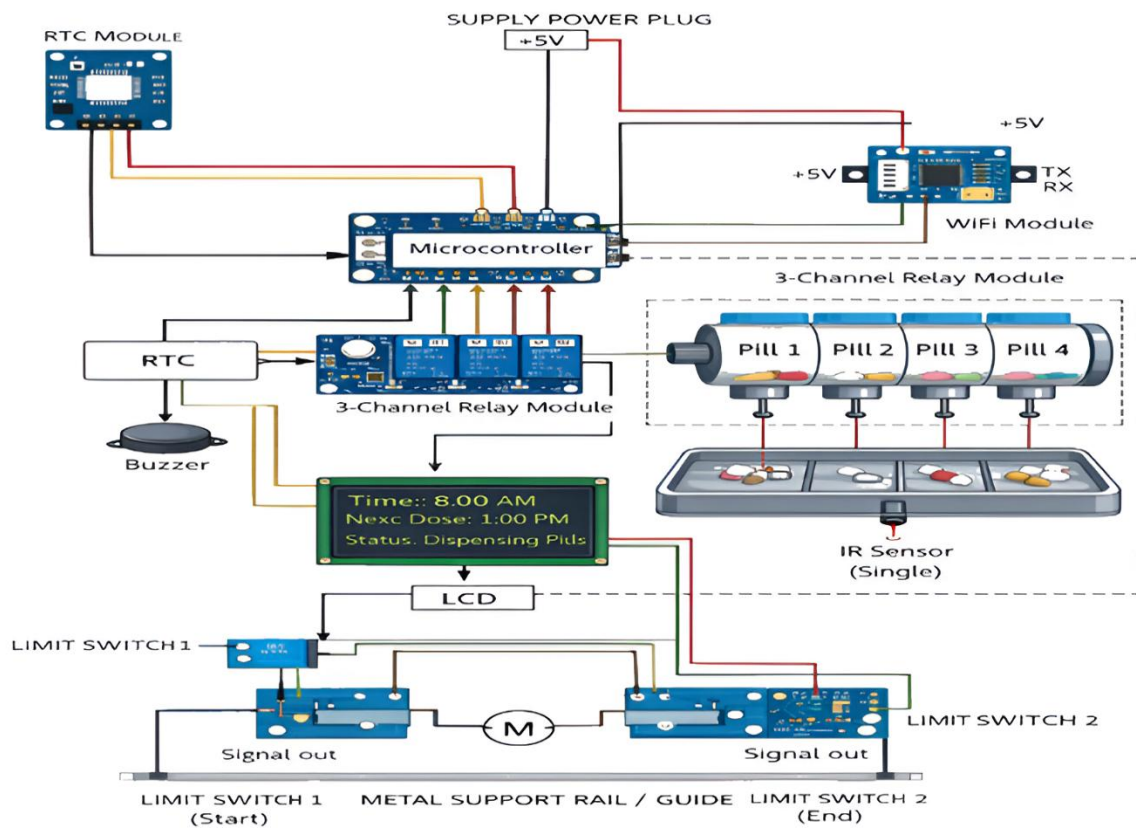


Figure 4.2.1 Circuit Diagram

The circuit diagram represents the complete working of the Smart Automated Medicine Dispensing and Alert System, showing how different electronic and mechanical components are interconnected to perform automatic medicine dispensing. At the center of the system is the microcontroller (Arduino/Node MCU), which acts as the main control unit. It receives inputs, processes data, and controls all output devices. The system is powered by a 5V power supply, which provides the required voltage to all components including the microcontroller, Wi-Fi module, relay module, and sensors.

At the center of the system is the microcontroller (Arduino/Node MCU), which acts as the main control unit. It receives inputs, processes data, and controls all output devices. The system is powered by a 5V power supply, which provides the required voltage to all components including the microcontroller, Wi-Fi module, relay module, and sensors.

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The RTC (Real-Time Clock) module is connected to the microcontroller to maintain accurate time. It helps the system trigger medicine dispensing at the correct scheduled time. The Wi-Fi module enables IoT communication, allowing the system to send notifications to caregivers and receive schedule data from the mobile application. The RTC (Real-Time Clock) module is connected to the microcontroller to maintain accurate time. It helps the system trigger medicine dispensing at the correct scheduled time. The Wi-Fi module enables IoT communication, allowing the system to send notifications to caregivers and receive schedule data from the mobile application.

A 3-channel relay module is used to control the dispensing mechanism. Each relay corresponds to a specific medicine compartment (Pill 1, Pill 2, Pill 3, Pill 4). When activated, the relay drives the motor to release the tablet from the selected compartment. The IR sensor is placed near the output tray to detect whether the medicine has been taken. This feedback is sent to the microcontroller to determine the medicine status. A buzzer is connected to provide an audio alert when it is time to take medicine, and an LCD display shows information such as current time, next dose, and system status. Additionally, limit switches are used to control the movement of the motor and ensure proper positioning of the dispensing mechanism along the rail. The DC motor drives the mechanical system to align and release the tablets.

Overall, the circuit integrates timing, control, sensing, alerting, and communication modules to create a fully automated and reliable medicine dispensing system with real-time monitoring. Additionally, limit switches are used to control the movement of the motor and ensure proper positioning of the dispensing mechanism along the rail. The DC motor drives the mechanical system to align and release the tablets.

4.3 WORKING OF PROJECT MODEL

The Smart Automated Medicine Dispensing and Alert System works by combining automation, sensing, and IoT communication to ensure timely medication. The system begins with storing tablets in a cylindrical container that is divided into multiple compartments for different time schedules such as morning, afternoon, and night. The user or caregiver sets the medication timings through a mobile application, and this schedule is stored in the microcontroller via the IoT module.

The microcontroller continuously monitors the current time and compares it with the stored schedule. When the scheduled time is reached, it activates the servo motor, which rotates the cylindrical storage to align the correct compartment. As a result, the required tablet is automatically dispensed into the output tray. Immediately after dispensing, a buzzer alert is generated to notify the user. The alert continues until the medicine is taken. Sensors placed near the dispensing area detect whether the tablet has been removed. Based on this detection, the system determines whether the medicine has been taken or not.

The IoT module then sends real-time notifications to the caregiver through the mobile application. If the medicine is taken, a confirmation message is sent; if not, an alert indicating a missed dose is transmitted. This ensures continuous monitoring and improves patient safety. Overall, the system provides a fully automated, reliable, and user-friendly solution that reduces human errors, improves medication adherence, and supports independent living for elderly and visually impaired individuals.

4.3.1 Medicine Storage System

Tablets are stored in a cylindrical container with multiple compartments for different time schedules such as morning, afternoon, and night. Each compartment holds the required medicine for a specific time.

4.3.2 Time Scheduling through Mobile App

The user or caregiver sets the medicine timings using a mobile application. These timings are sent to the microcontroller through the IoT module and stored in the system.

4.3.3 Microcontroller Control Unit

The microcontroller (ESP32/Node MCU) acts as the main controller. It continuously monitors the current time and compares it with the scheduled timings to control the dispensing process.

4.3.4 Automatic Tablet Dispensing

When the scheduled time is reached, the microcontroller activates the servo motor. The motor rotates the cylindrical storage to align the correct compartment, allowing the tablet to fall into the output tray.

4.3.5 Buzzer Alert System

A buzzer is activated immediately after dispensing the tablet to alert the user. The alert continues until the medicine is taken, ensuring that the user does not miss the dose.

4.3.6 Medicine Detection System

Sensors such as IR sensors or switches are placed near the dispensing area to detect whether the tablet has been taken. This helps the system to confirm medicine intake.

4.3.7 Notification and Alert System

Based on the sensor input, the system sends real-time notifications to the caregiver through the IoT module. If the medicine is taken, a confirmation message

is sent. If the medicine is not taken within the specified time, an alert message is sent to indicate a missed dose.

4.3.8 IoT Connectivity and Remote Monitoring

The system uses Wi-Fi connectivity to communicate with a mobile application. Caregivers can monitor medication schedules, intake status, and alerts remotely. This improves supervision and ensures patient safety even from a distance.

4.3.9 Data Storage and Tracking

The system stores information such as medication time, taken status, and missed doses. This data can be accessed through the mobile app for tracking and analysis, helping caregivers understand the patient's medication routine.

4.3.10 Power Supply and System Operation

The system is powered using a stable power supply that ensures continuous operation of all components such as the microcontroller, sensors, and motor. Proper power management helps maintain reliability and smooth functioning.

4.3.11 User Interface and Control

The mobile application provides an easy-to-use interface for setting schedules, viewing alerts, and monitoring system status. It simplifies operation and makes the system accessible to users of all age groups. The system uses Wi-Fi connectivity to communicate with a mobile application. Caregivers can monitor medication schedules, intake status, and alerts remotely. This improves supervision and ensures patient safety even from a distance.

CHAPTER 5

HARDWARE IMPLEMENTATION

5.1 IMPLEMENTATION OF HARDWARE

The given images represent the practical implementation of the Smart Automated Medicine Dispensing and Alert System, consisting of both electronic and mechanical sections. In the control circuit section, the Arduino UNO acts as the main controller, coordinating all operations of the system. It is connected to multiple relay modules, which function as electronic switches to control the DC motors used for dispensing tablets.

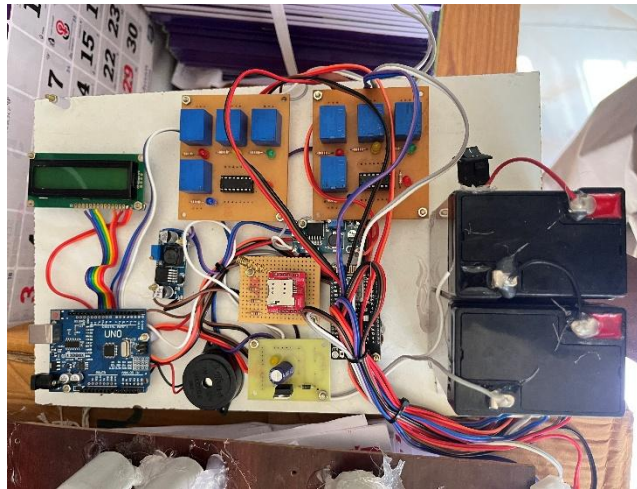


Figure 5.1.1 Prototype of the System

The given images represent the practical implementation of the Smart Automated Medicine Dispensing and Alert System, consisting of both electronic and mechanical sections. In the control circuit section, the Arduino UNO acts as the main controller, coordinating all operations of the system. It is connected to multiple relay modules, which function as electronic switches to control the DC motors used for dispensing tablets. An LCD display is included to show information such as current time, next dosage, and system status, while a buzzer provides an audio alert to remind the user to take the medicine. The entire system is powered by a suitable power supply and battery setup, and all components are interconnected using proper wiring.

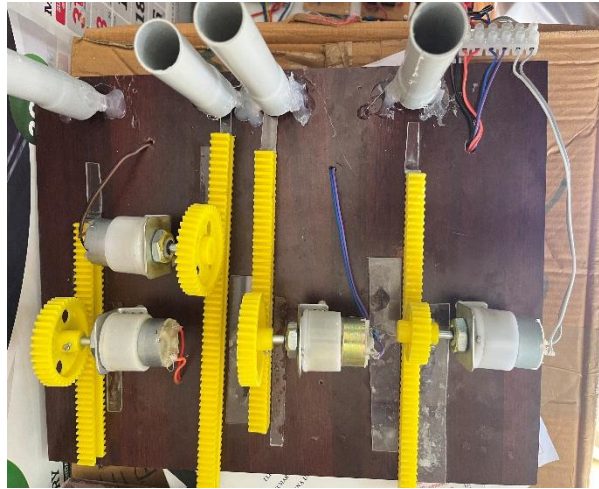


Figure 5.1.2 Tablet Storage

In the mechanical section, DC motors are connected to a rack and pinion gear mechanism, which converts rotational motion into linear motion. Tablets are stored in vertical tubes, and each tube corresponds to a specific time schedule. When the microcontroller activates a relay at the scheduled time, the corresponding motor rotates, moving the gear mechanism and allowing the tablet to fall from the tube into the collection area.

5.2 COMPONENTS AND DESCRIPTION

ESP32/Node MCU Microcontroller



Figure5.2.1 ESP32/ NodeMCU Microcontroller

This acts as the main controller of the system. It processes the timing schedule, controls the servo motor, reads sensor data, and manages communication with the mobile application through IoT.

Cylindrical Medicine StorageBox

A rotating cylindrical container with multiple compartments used to store tablets for different time schedules such as morning, afternoon, and night. It ensures organized storage and accurate dispensing.

Servo Motor



Figure5.2.2 SERVO MOTOR

The servo motor is used to rotate the cylindrical storage. It moves to a specific position at the scheduled time to release the correct tablet automatically.

IRSensor/Limit Switch



Figure5.2.3 IR SENSOR

These sensors are used to detect whether the tablet has been taken from the dispensing area. They provide feedback to the system for monitoring medicine intake.

Buzzer



Figure5.2.4 BUZZER

The buzzer provides an audio alert when it is time to take medicine. It continues to sound until the tablet is taken, ensuring the user is notified properly.

Mobile Application

Used by users or caregivers to set medicine schedules, monitor system status, and receive notifications about medicine intake.

5.3HARDWARE COMPONENTS USED

Table 5.3.1 Hardware Components

S. No	Components Used	Quantity
1.	ESP32 / NODE MCU MICROCONTROLLER	1
2.	CYLINDRICAL MEDICINE STORAGE BOX	1
3.	SERVO MOTOR	1
4.	IR Sensor / Limit Switch	1
5.	Buzzer	1
6.	Shunt Resistance Module	1
7.	Relay Module	1
8.	Lead Acid Batteries	1
9.	LED	1

CHAPTER 6

RESULT AND DISCUSSION

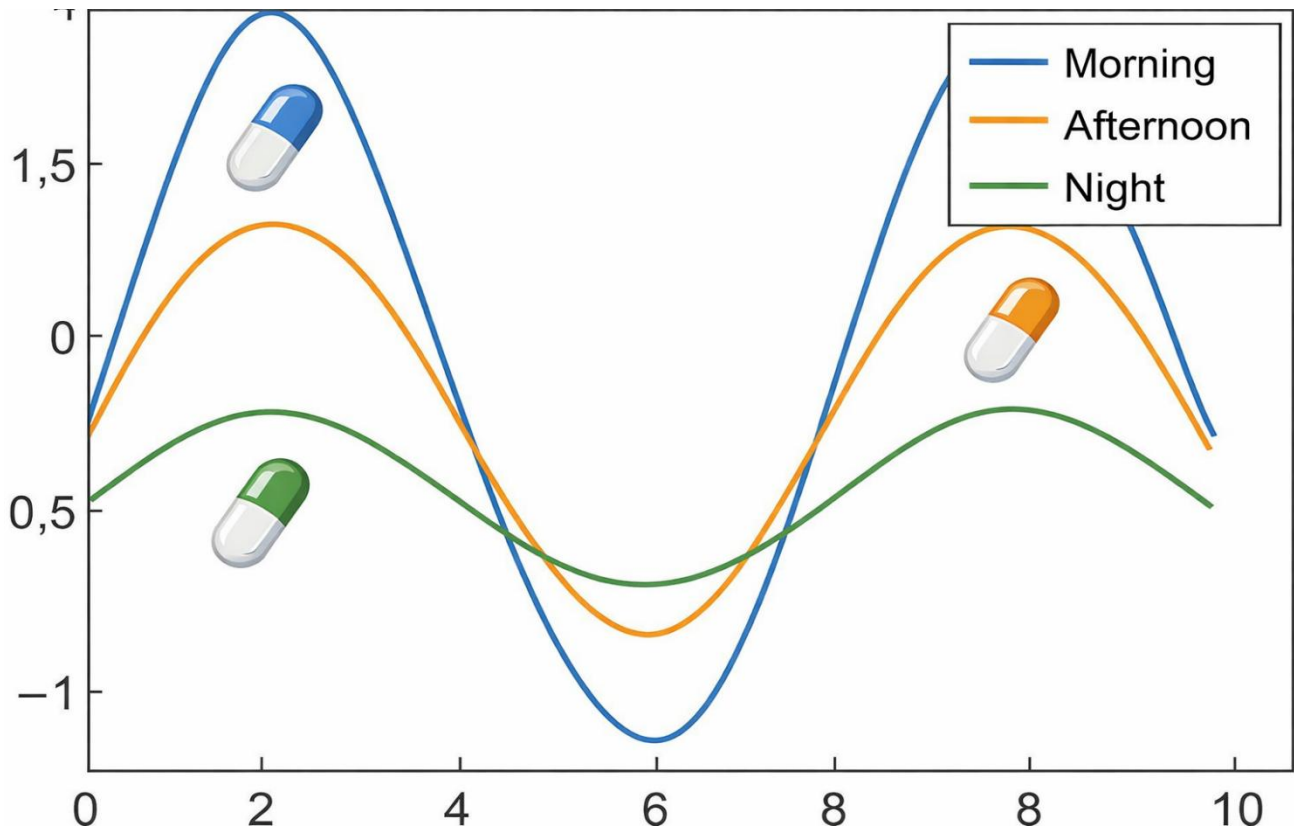
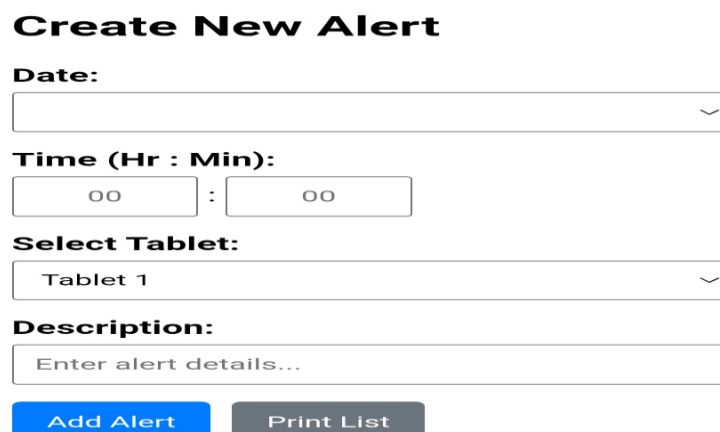


Figure 6.1 Simulated Power Flow Graph

The Smart Automated Medicine Dispensing and Alert System was successfully developed and tested to evaluate its performance and reliability. The system was able to dispense tablets accurately at the scheduled times using the servo motor and cylindrical storage mechanism. The timing control worked effectively, ensuring that medicines were released without delay. The buzzer alert responded immediately, helping users to take medicine on time.

The sensor system accurately detected whether the tablet was taken or not, providing reliable feedback to the system. Based on this, notifications were successfully sent to caregivers through the IoT module, confirming medicine intake or indicating missed doses. The communication between the device and mobile application was stable, enabling real-time monitoring.

During testing, the system performed consistently under different conditions and showed good reliability for continuous use. It reduced the chances of missed doses and improved medication adherence. The system also minimized the need for manual supervision, making it highly useful for elderly and visually impaired individuals. Overall, the project demonstrated effective integration of automation and IoT technology, providing a practical and user-friendly solution for better medication management and healthcare support. Overall, the project demonstrated effective integration of automation and IoT technology, providing a practical and user-friendly solution for better medication management and healthcare support.



Create New Alert

Date:

Time (Hr : Min):
 :

Select Tablet:

Description:

Figure 6.2 User Interface for Medication Scheduling and Alert Configuration

If the battery capacity reaches 40% or less, the device will automatically activate a low-energy protection mode. In this case, the control unit is only allowed to connect the low-priority load, while the medium- and high-priority.

The Smart Automated Medicine Dispensing and Alert System is an innovative healthcare solution designed to help elderly and visually impaired individuals manage their medication effectively. The system uses IoT technology to automate the process of medicine dispensing based on predefined schedules. Medicines are stored in a cylindrical container with separate compartments, and a servo motor is used to rotate and release the correct tablet at the right time.

A buzzer alert is provided to remind the user to take the medicine, ensuring that doses are not missed. The system also includes sensors to detect whether the medicine has been taken or not. Based on this information, real-time notifications are sent to caregivers through a mobile application. This feature allows remote monitoring and ensures patient safety even when caregivers are not physically present. The project focuses on reducing human errors, improving medication adherence, and promoting independent living. It is designed to be simple, cost-effective, and user-friendly, making it suitable for home healthcare applications. The project focuses on reducing human errors, improving medication adherence, and promoting independent living. It is designed to be simple, cost-effective, and user-friendly, making it suitable for home healthcare applications. By combining automation, sensing, and IoT communication, the system provides a reliable solution for better health management.

This image shows a mobile or web application interface used in the Smart Automated Medicine Dispensing and Alert System to schedule medication reminders. The screen is titled “Create New Alert”, which allows the user or caregiver to set a new medicine alert easily. This image shows a mobile or web application interface used in the Smart Automated Medicine Dispensing and Alert System to schedule medication reminders. The screen is titled “Create New Alert”, which allows the user or caregiver to set a new medicine alert easily.

At the top, there is a Date field, where the user selects the specific day for the medication. Below that, the Time (Hr : Min) section allows the user to set the exact time for taking the medicine by entering hours and minutes. The Select Tablet option is a dropdown menu where the user chooses which tablet or compartment should be dispensed at that time. There is also a Description field, where additional details about the medicine or instructions can be entered. At the bottom, two buttons are provided: “Add Alert” to save the scheduled reminder and “Print List” to view or print all saved medication schedules.

Overall, this interface is user-friendly and helps in easily setting and managing medicine timings, ensuring accurate scheduling and better medication adherence. . At the bottom, two buttons are provided: “Add Alert” to save the scheduled reminder and “Print List” to view or print all saved medication schedules.

ID	Target	Date & Time	Description	Status	Action
1	Tablet 1	2026-04-03 18:05:00	test	MISSED	Delete
3	Tablet 1	2026-04-04 18:13:00	rr	COMPLETED	Delete
6	Tablet 1	2026-04-06 10:24:00	Tablet	MISSED	Delete
7	Tablet 1	2026-04-06 10:26:00	Tablet	MISSED	Delete

Figure 6.3 Medication History and Alert Status Dashboard

This image shows a medicine alert history dashboard from the Smart Automated Medicine Dispensing and Alert System. It displays a table that helps users or caregivers track medication schedules and their status. This image shows a medicine alert history dashboard from the Smart Automated Medicine Dispensing and Alert System. It displays a table that helps users or caregivers track medication schedules and their status. The table contains several columns such as ID, Target, Date & Time, Description, Status, and Action. Each row represents a scheduled medicine alert. The Target column indicates the type of medicine (tablet), while the Date & Time column shows when the medicine was supposed to be taken. The Description field provides additional details about the medicine.

The most important part is the Status column, which clearly shows whether the medicine was “COMPLETED” (taken successfully) or “MISSED” (not taken on time). This helps caregivers quickly understand the patient’s medication adherence. The Action column includes a “Delete” button, allowing users to remove specific entries from the list if needed. The most important part is the Status column, which clearly shows whether the medicine was “COMPLETED” (taken successfully)

or “MISSED” (not taken on time). This helps caregivers quickly understand the patient’s medication adherence. The Action column includes a “Delete” button, allowing users to remove specific entries from the list if needed. Overall, this interface provides a clear and organized way to monitor medication history, identify missed doses, and ensure better healthcare management through real-time tracking.

This image shows a medicine alert history dashboard from the Smart Automated Medicine Dispensing and Alert System. It displays a table that helps users or caregivers track medication schedules and their status. This image shows a medicine alert history dashboard from the Smart Automated Medicine Dispensing and Alert System. It displays a table that helps users or caregivers track medication schedules and their status. The table contains several columns such as ID, Target, Date & Time, Description, Status, and Action. Each row represents a scheduled medicine alert. The Target column indicates the type of medicine (tablet), while the Date & Time column shows when the medicine was supposed to be taken. The Description field provides additional details about the medicine. This image shows a medicine alert history dashboard from the Smart Automated Medicine Dispensing and Alert System. It displays a table that helps users or caregivers track medication schedules and their status. This image shows a medicine alert history dashboard from the Smart Automated Medicine Dispensing and Alert System. It displays a table that helps users or caregivers track medication schedules and their status. The table contains several columns such as ID, Target, Date & Time, Description, Status, and Action. Each row represents a scheduled medicine alert. The Target column indicates the type of medicine (tablet), while the Date & Time column shows when the medicine was supposed to be taken. The Description field provides additional details about the medicine.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The Smart Automated Medicine Dispensing and Alert System provides an effective and reliable solution for managing daily medication schedules, particularly for elderly and visually impaired individuals who often face difficulties in remembering and handling medicines. The system successfully automates the process of medicine dispensing using a cylindrical storage mechanism controlled by a servo motor, ensuring that the correct medicine is delivered at the right time without human intervention. This significantly reduces the chances of missed doses, incorrect timing, and medication errors.

The inclusion of a buzzer alert system ensures that users are clearly notified when it is time to take their medicine, improving adherence to prescribed schedules. Additionally, the integration of sensors enables the system to detect whether the medicine has been taken, providing accurate feedback. The IoT-based communication feature enhances the system further by sending real-time notifications to caregivers about the medication status, whether taken or missed. This ensures continuous monitoring and increases patient safety, even when caregivers are not physically present. The system has demonstrated good reliability, efficiency, and ease of use during implementation. It is designed to be cost-effective and user-friendly, making it suitable for home healthcare applications. By reducing dependency on caregivers and supporting independent living, the system improves the quality of life for users.

The system has demonstrated good reliability, efficiency, and ease of use during implementation. It is designed to be cost-effective and user-friendly, making it suitable for home healthcare applications. By reducing dependency on caregivers and supporting independent living, the system improves the quality of life for users. Overall, this project highlights the importance of combining automation and IoT

in healthcare systems. It provides a practical, scalable, and intelligent solution for improving medication adherence, enhancing patient safety, and supporting better healthcare management in modern society.

7.2 FUTURE SCOPE

The Smart Automated Medicine Dispensing and Alert System can be further enhanced by incorporating advanced features to improve its functionality, reliability, and user experience. Future developments may include the integration of voice assistance to guide visually impaired users through audio instructions, making the system more accessible. The addition of medicine level detection sensors can help monitor the availability of tablets and provide automatic refill alerts. The system can also be upgraded with a battery backup or rechargeable power system to ensure uninterrupted operation during power failures. Furthermore, integrating cloud storage and data analytics can enable long-term tracking of medication history, which can be useful for doctors and caregivers. The development of a more advanced mobile application with an improved user interface and flexible scheduling options can enhance usability. In addition, implementing AI-based features can help predict missed doses and provide smart reminders. Overall, these enhancements can transform the system into a more intelligent, efficient, and comprehensive healthcare solution.

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APPENDIX

```
#include <Wire.h>
#include <EEPROM.h>
#include <RTCLib.h>
#include <LiquidCrystal.h>
#include <Servo.h>

LiquidCrystal lcd(3,4,5,6,7,8);
RTC_DS1307 RTC;

Servo myServo;

// Pins
int next = A0;
int INC = A1;
int set_mad = A2;

#define buzzer A6
#define LED 9
#define SERVO_PIN 10

int HOUR, MINUT, SECOND;
int hours1, minut, add = 11;

bool medicineTaken = false;
unsigned long alertStartTime = 0;
```

```

// ----- SETUP -----

void setup() {
  Wire.begin();
  RTC.begin();
  lcd.begin(16, 2);

  pinMode(INC, INPUT);
  pinMode(next, INPUT);
  pinMode(set_mad, INPUT);

  pinMode(buzzer, OUTPUT);
  pinMode(LED, OUTPUT);

  myServo.attach(SERVO_PIN);

  Serial.begin(9600);

  lcd.setCursor(0, 0);
  lcd.print("Medicine Reminder");
  lcd.setCursor(0, 1);
  lcd.print("Using Arduino");
  delay(2000);

  if (!RTC.isrunning()) {
    RTC.adjust(DateTime(__DATE__, __TIME__));
  }
}

```

```

// ----- LOOP -----
void loop() {
    DateTime now = RTC.now();

    // SET TIME MODE
    if (digitalRead(set_mad) == 0) {
        lcd.clear();
        lcd.print("Set Medicine Time");
        delay(2000);

        lcd.clear();
        lcd.print("Enter Time 1");
        default();
        time(1);

        lcd.clear();
        lcd.print("Enter Time 2");
        default();
        time(2);

        lcd.clear();
        lcd.print("Enter Time 3");
        default();
        time(3);

        lcd.clear();
        lcd.print("Time Set Done");
        delay(2000);
    }
}

```

```

}

// DISPLAY TIME
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Time:");
lcd.print(now.hour());
lcd.print(":");
lcd.print(now.minute());
lcd.print(":");
lcd.print(now.second());

lcd.setCursor(0, 1);
lcd.print("Date:");
lcd.print(now.day());
lcd.print("/");
lcd.print(now.month());
lcd.print("/");
lcd.print(now.year());

HOUR = now.hour();
MINUT = now.minute();
SECOND = now.second();

match();

delay(500);
}

```

```

// ----- DEFAULT DISPLAY -----
void default() {
    lcd.setCursor(0, 1);
    lcd.print(HOUR);
    lcd.print(":");
    lcd.print(MINUT);
    lcd.print(":");
    lcd.print(SECOND);
}

// ----- TIME SET FUNCTION -----
void time(int x) {
    int temp = 1;

    while (temp == 1) {
        if (digitalRead(INC) == 0) {
            HOUR++;
            if (HOUR == 24) HOUR = 0;
            delay(200);
        }

        lcd.clear();
        lcd.print("Set Hour ");
        lcd.print(x);
        lcd.setCursor(0, 1);
        lcd.print(HOUR);
    }
}

```

```

if (digitalRead(next) == 0) {
    EEPROM.write(add++, HOUR);
    temp = 2;
    delay(200);
}
}

while (temp == 2) {
    if (digitalRead(INC) == 0) {
        MINUT++;
        if (MINUT == 60) MINUT = 0;
        delay(200);
    }

    lcd.clear();
    lcd.print("Set Minute ");
    lcd.print(x);
    lcd.setCursor(0, 1);
    lcd.print(MINUT);

    if (digitalRead(next) == 0) {
        EEPROM.write(add++, MINUT);
        temp = 0;
        delay(200);
    }
}
}

```

```

// ----- MATCH FUNCTION -----

void match() {
    int tem[17];

    for (int i = 11; i < 17; i++) {
        tem[i] = EEPROM.read(i);
    }

    if (HOUR == tem[11] && MINUT == tem[12]) {
        triggerMedicine("Group 1");
    }

    if (HOUR == tem[13] && MINUT == tem[14]) {
        triggerMedicine("Group 2");
    }

    if (HOUR == tem[15] && MINUT == tem[16]) {
        triggerMedicine("Group 3");
    }
}

// ----- MEDICINE ACTION -----

void triggerMedicine(String groupName) {
    lcd.clear();
    lcd.print("Take " + groupName);
    lcd.setCursor(0, 1);
    lcd.print("Medicine");
}

```

```

// Servo rotate
myServo.write(90);
delay(1000);
myServo.write(0);

digitalWrite(LED, HIGH);

alertStartTime = millis();
medicineTaken = false;

Serial.println("Medicine Time: " + groupName);

for (int i = 0; i < 5; i++) {
    beep();
}

checkMedicineTaken();
}

// ----- CHECK TAKEN -----
void checkMedicineTaken() {
    while (millis() - alertStartTime < 10000) {
        if (digitalRead(next) == 0) {
            medicineTaken = true;
            break;
        }
    }
}

```

```

lcd.clear();

if (medicineTaken) {
    lcd.print("Taken");
    Serial.println("Taken");
} else {
    lcd.print("Missed Dose");
    Serial.println("Missed");
}

digitalWrite(LED, LOW);
delay(2000);
}

// ----- BUZZER -----
void beep() {
    digitalWrite(buzzer, HIGH);
    delay(300);
    digitalWrite(buzzer, LOW);
    delay(300);
}

```

